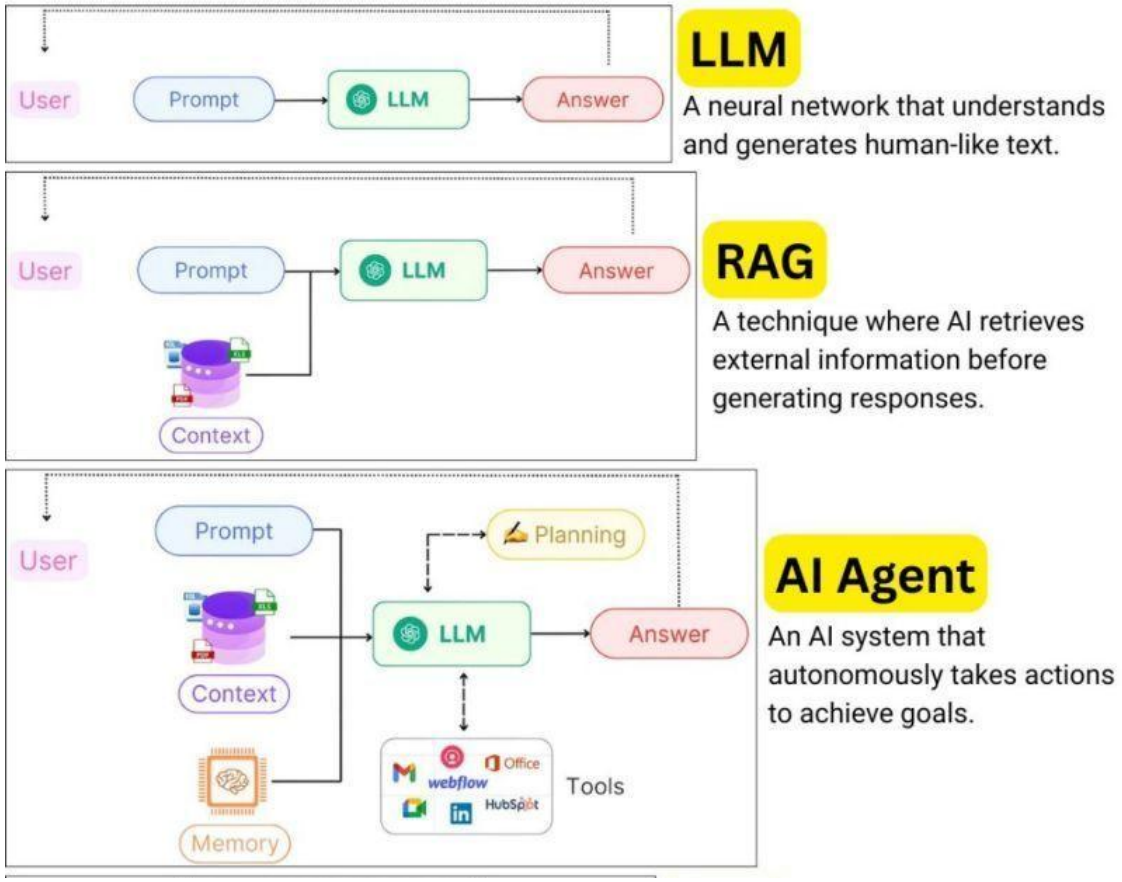
Retrieval-Augmented Generation

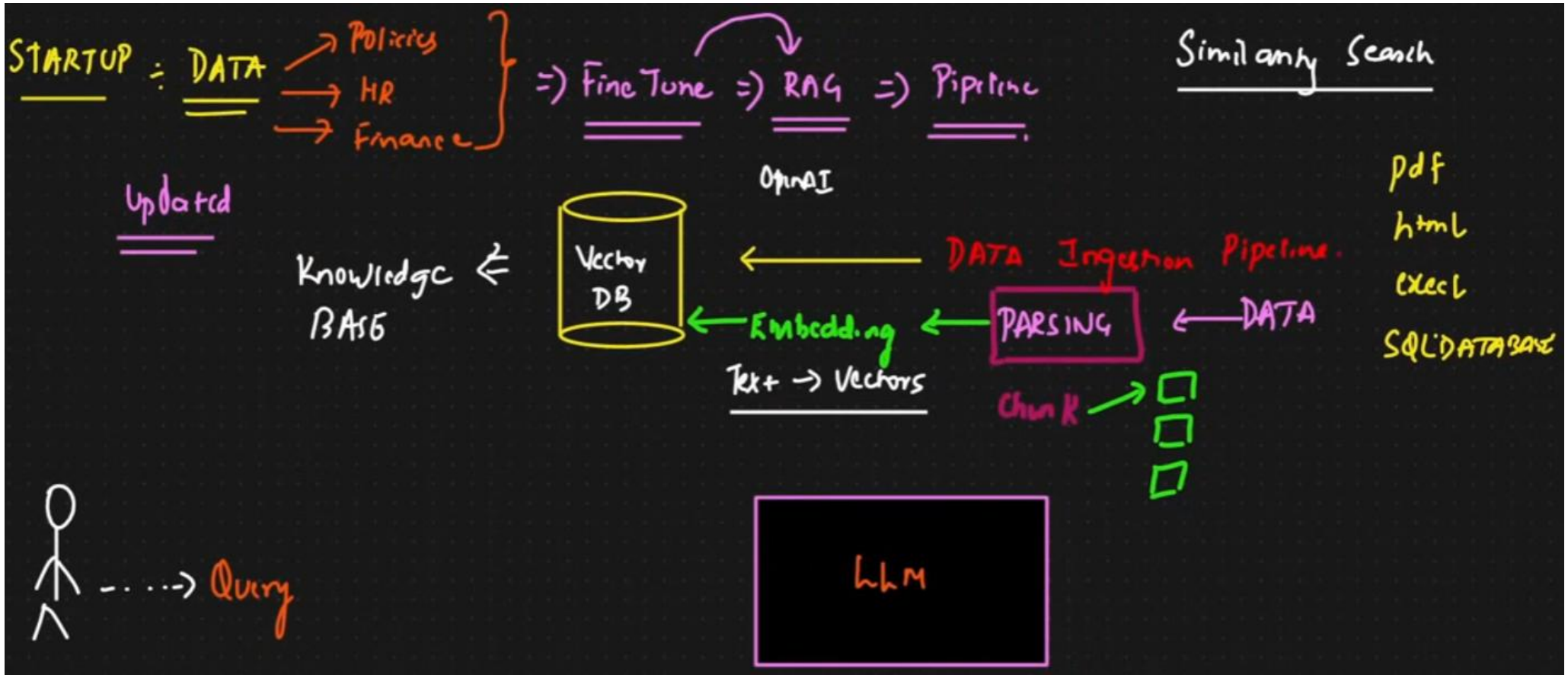
RAG stands for Retrieval-Augmented Generation. It's a way to make large language models more reliable by giving them access to external knowledge before they generate an answer.

Instead of relying solely on what the model learned during training—which can be outdated or inaccurate—RAG first retrieves relevant information from a knowledge source, like a database, a set of documents, or a vector store. Then, it feeds that information to the model as context, so the model's response is grounded in real, up-to-date data.

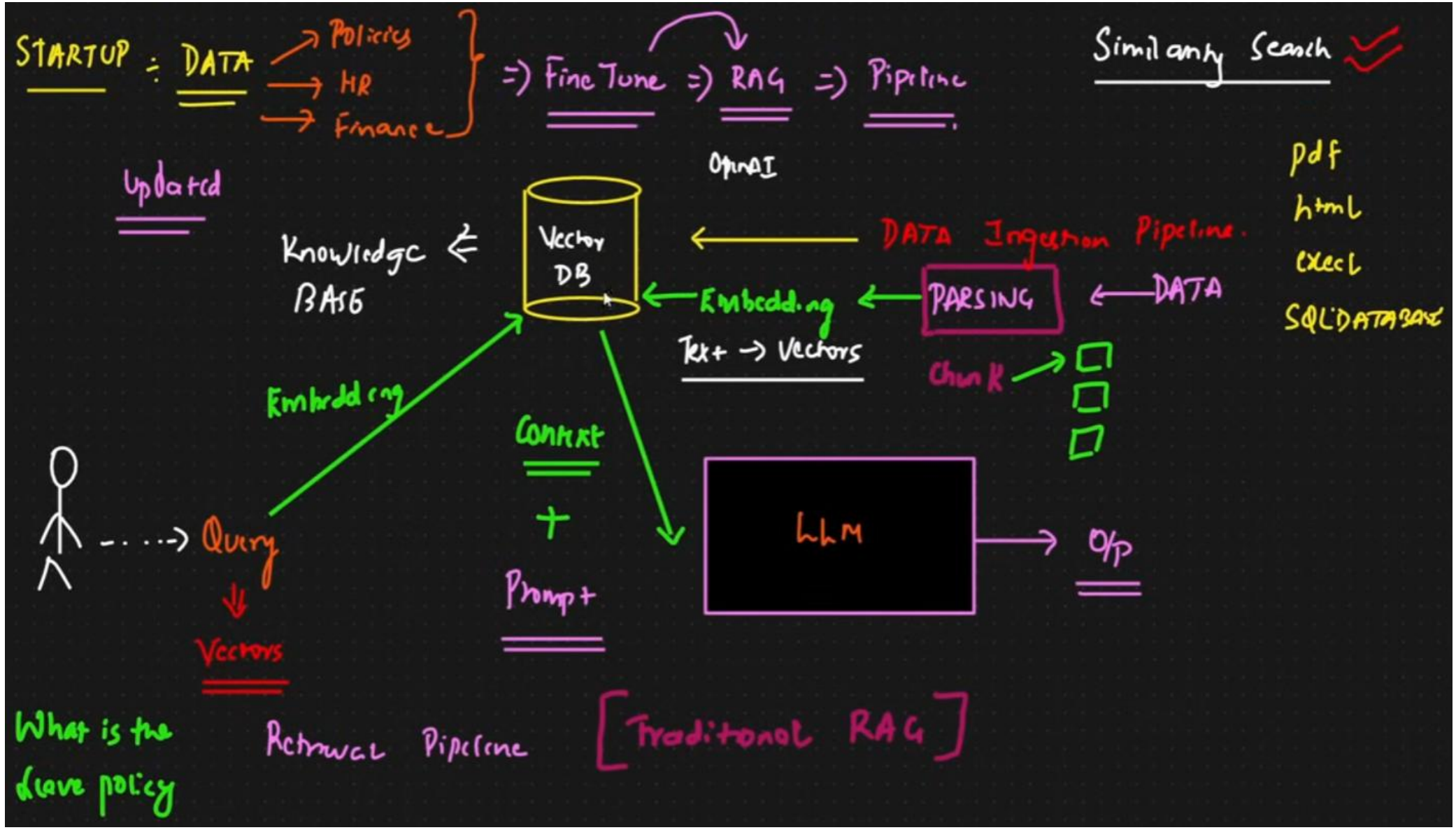
What this does is it significantly reduces hallucinations—those confident but wrong answers—and ensures the output is accurate, traceable, and current. It's a core technique in building trustworthy AI applications, especially for things like customer support, internal knowledge bases, or any scenario where you need the model to stick to the facts.

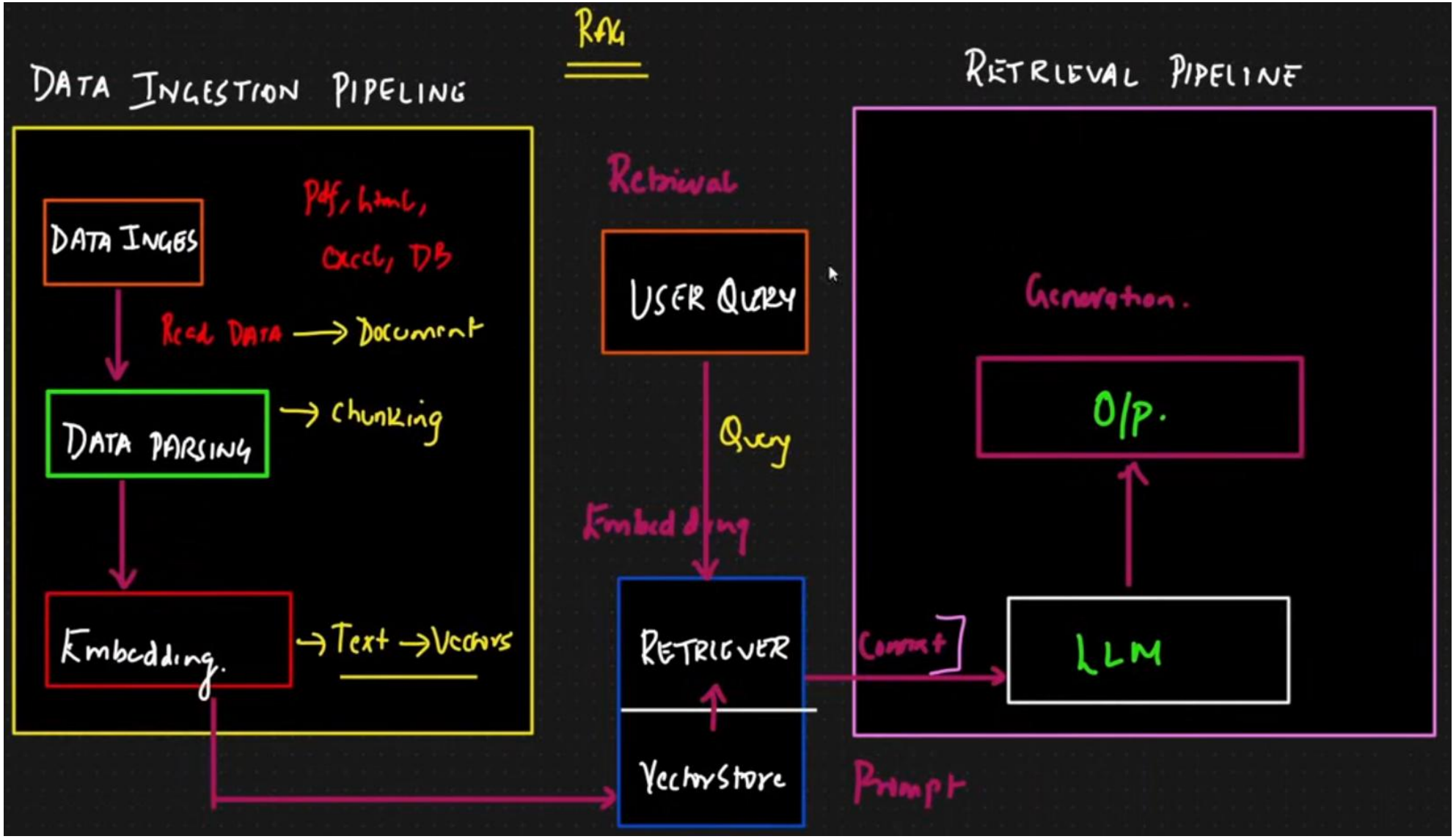


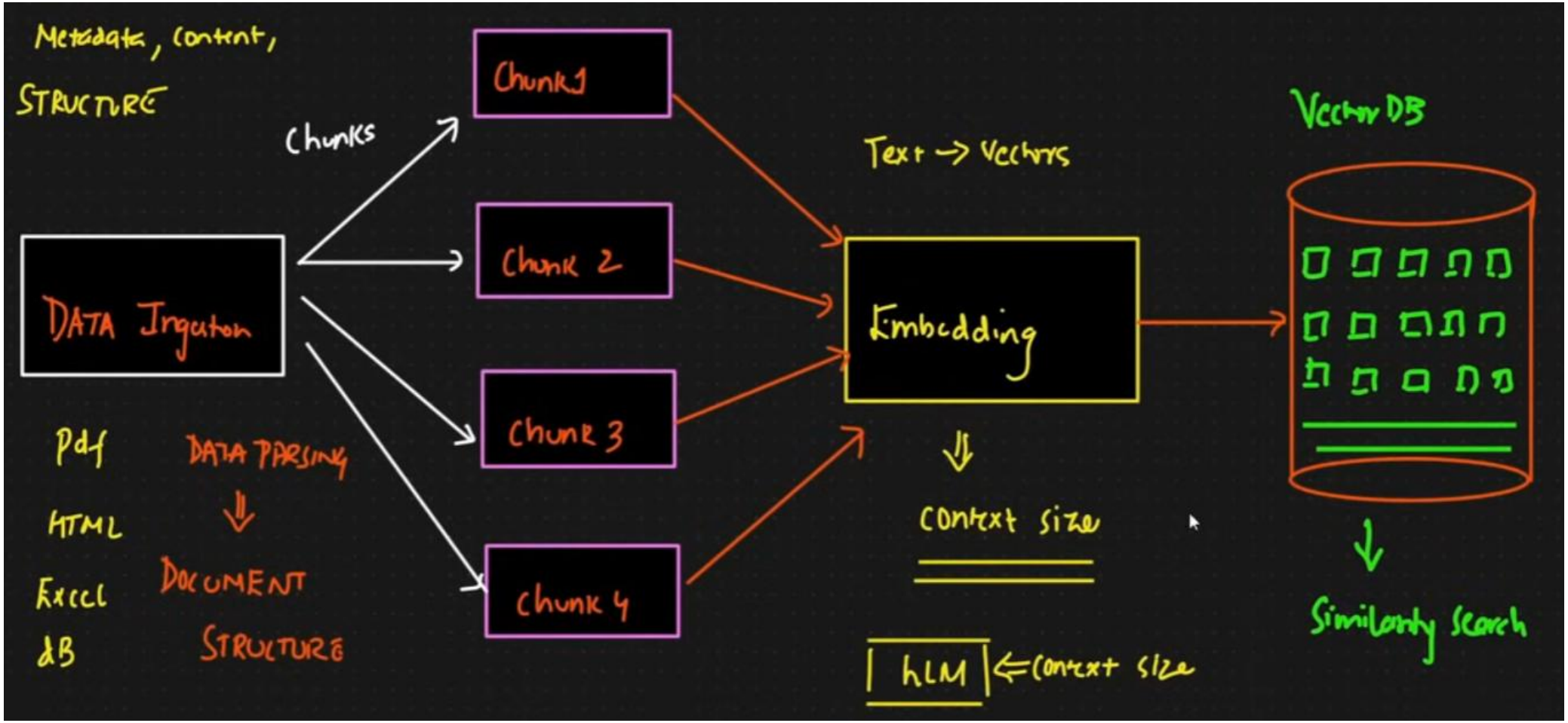
data ingestion pipeline



retrieval pipeline




RAG
DATA INGESTION PIPELING
DATA INGES
PDF, html,
excel, DB
Read DATA → Document
DATA PARSING
→ chunking
Embedding.
→ Text → Vectors
Retrieval
USER QUERY
Query
Embedding
RETRIEVER
VectorStore
Context
Prompt
RETRIEVAL PIPELINE
Generation.
O/P.
LLM

Agenda

1) Document Structure
2) Complete RAG Pipeline
   - → DATA INGESTION PIPELINE
   - → QUERY RETRIEVAL PIPELINE